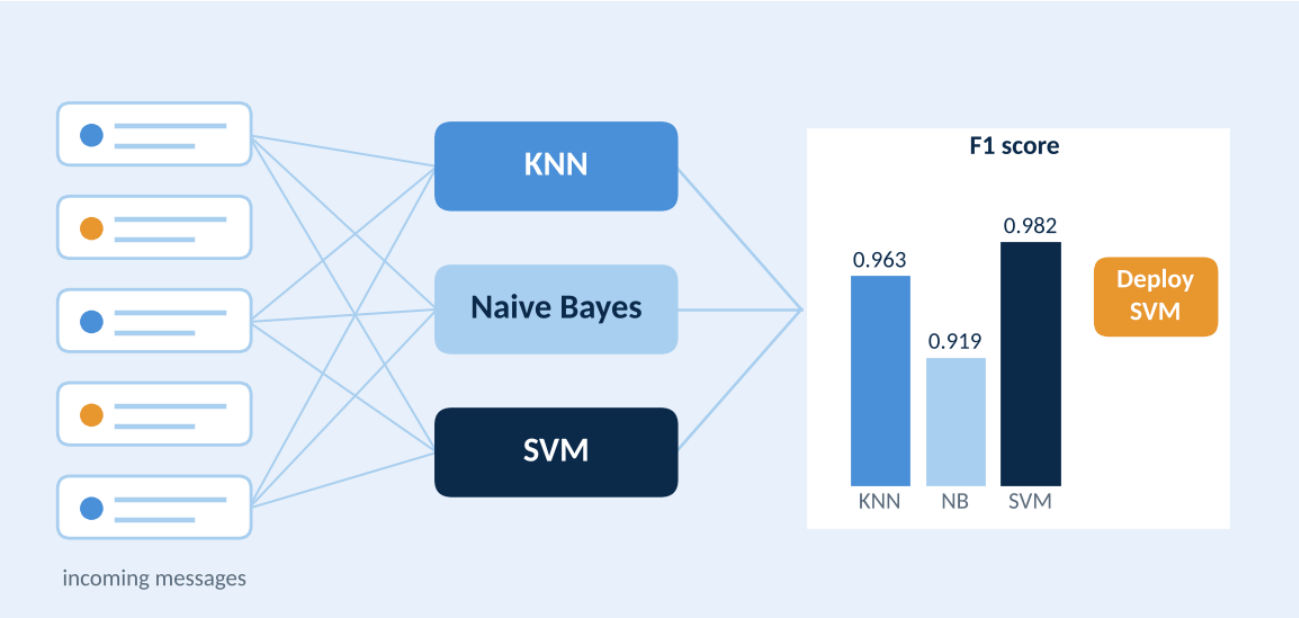
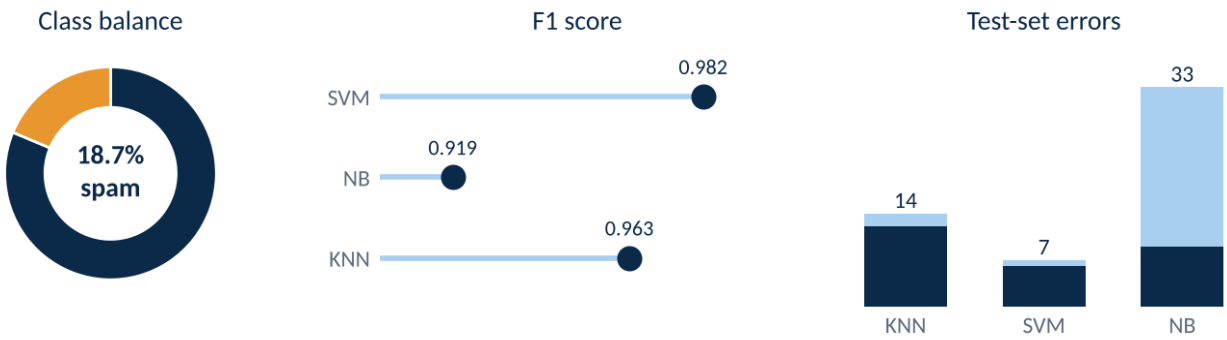
SUPERVISED MACHINE LEARNING

MESSAGE

INTELLIGENCE SYSTEM

Part G: Final Analysis & Reporting





Part G: Final Analysis & Reporting

Prepare a final report summarizing:

23.1 Strengths and weaknesses of each classifier

	KNN (K = 5)	SVM (RBF)	Naive Bayes (Gaussian)
Strengths	Accuracy 98.65%, F1 96.32%. Simple, no training step; stable for K = 3–9 and across Euclidean, Manhattan and Minkowski distances; only 2 false alarms	Accuracy 99.33%, F1 98.18%, best on all four metrics. Margin-based boundary generalises well; the linear kernel needs only 83 support vectors (230 for RBF)	Accuracy 96.83%, F1 91.85%. Very fast and lightweight; gives class probabilities; second-best recall (95.4%)
Weaknesses	Lowest recall: 12 of 195 spam missed; must store all data, so prediction slows as data grows; needs scaling	RBF model is harder to explain; needs tuning of C and kernel; costlier to train on very large data	Lowest precision: 24 legitimate messages flagged; independence and Gaussian assumptions do not hold here

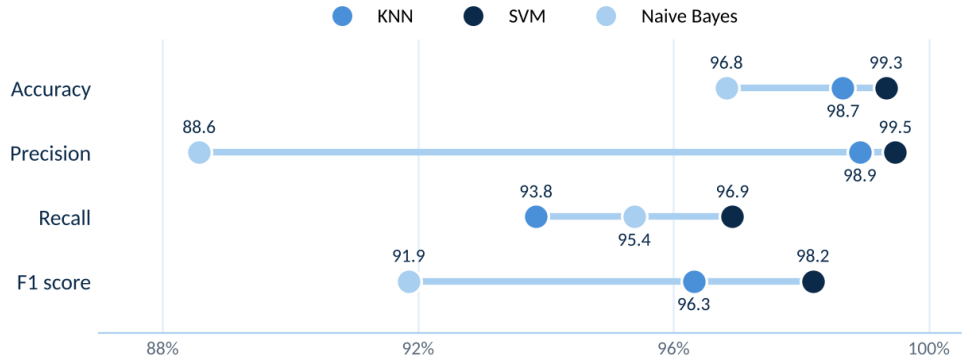


Fig 1. Test-set metrics by model (%; axis starts at 88%)

23.2 Impact of probability assumptions in Naive Bayes

Naive Bayes multiplies feature probabilities as if each feature were **independent given the class** and **Gaussian**. Neither holds here:

- **Correlated features:** num_urls with num_digits (0.71), message_length with num_urls (0.59) and num_urls with messages_sent_last_24h (0.55). Their evidence is counted twice.
- **Non-Gaussian features:** num_urls is only 0 or 1, and num_digits, hour_of_day and day_of_week are discrete.
- **Effect:** probabilities become over-confident. 92.8% of predictions are above 99% or below 1%, and mean confidence is 98.9% against 96.8% actual accuracy. Precision falls to 88.6% (24 false positives) versus 99.5% for SVM, although strong single signals (P(URL|Spam) = 0.739 vs 0.084) keep recall at 95.4%.

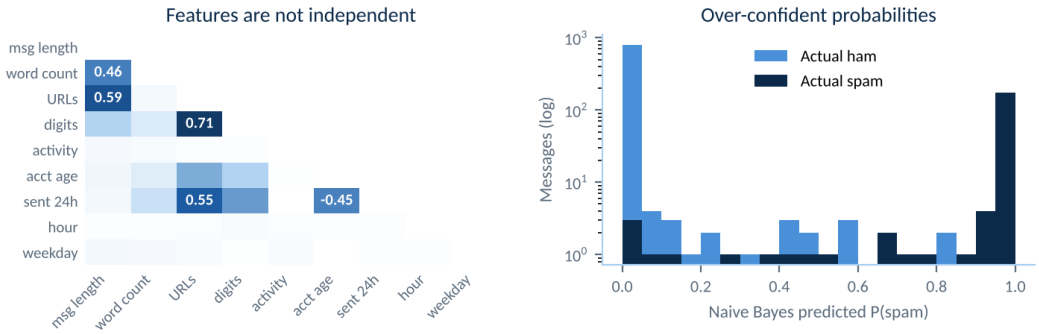


Fig 2. Feature correlations (left) and Naive Bayes confidence on the test set (right)

23.3 Trade-offs between interpretability and performance

- **Naive Bayes:** most transparent (priors and per-feature likelihoods can be read directly) but lowest F1 (91.85%).
- **KNN:** easy to explain by example ("similar to known spam") but has no compact model to inspect, and it misses the most spam.
- **SVM:** the RBF kernel is a black box (230 support vectors). The linear kernel reaches 99.42% accuracy with readable weights. The main drivers are URLs (+2.63), messages sent in 24h (+2.57) and account age (−2.32).

Takeaway: on this data the more interpretable linear SVM gives up nothing to the RBF kernel (1 message of 1,040 separates them), so complexity is not rewarded.

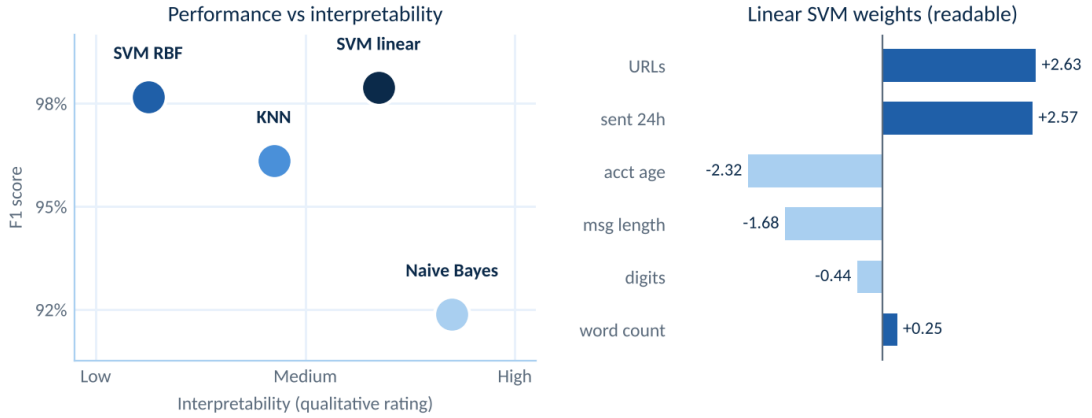


Fig 3. F1 against a qualitative interpretability rating, and linear SVM weights

23.4 Business recommendation for real-world deployment

- **Deploy SVM** (linear kernel as the default) as the main filter. It has the fewest errors in both directions: 1 legitimate message blocked and 5–6 spam missed out of 1,040.
- **Keep Naive Bayes** only as a fast first-pass or baseline, and do not rely on its raw probabilities without calibration.
- **Do not use KNN** in production: recall is lowest of the tuned models and prediction cost grows with data.
- **Reduce risk:** send messages near the decision margin to review, track both error types, retrain regularly, and run a pilot first. The dataset is highly repetitive and the keyword scores were excluded, so live traffic may be harder.

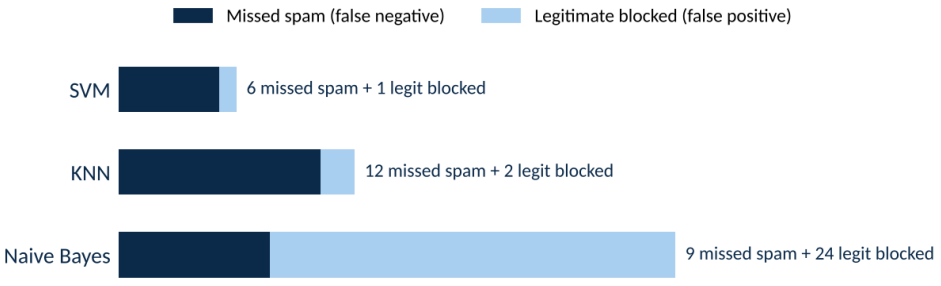


Fig 4. Errors on the 1,040 test messages

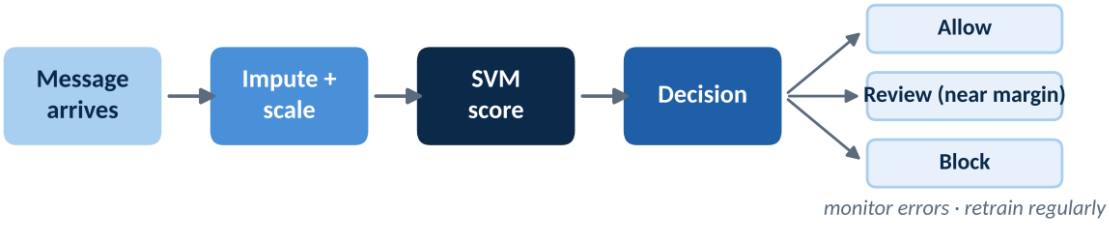


Fig 5. Suggested deployment flow